

Predictive Forecasting of Care Load & Placement Demand

EXECUTIVE SUMMARY

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1. Project Overview

The Predictive Forecasting of Care Load & Placement Demand project develops an end-to-end forecasting system to estimate future HHS Care load and short-term discharge demand using historical operational data. The project combines time-series analysis, statistical forecasting, machine-learning models, uncertainty estimation, and scenario analysis. The final results are presented through an interactive Streamlit dashboard designed to provide forward-looking analytical information for planning. The system is intended as a decision-support and planning tool rather than a replacement for operational judgment.

2. Problem Statement

The available operational dataset contains historical daily information about children entering, remaining in, and exiting HHS care. Historical reporting describes previous conditions but does not directly provide estimates of future care load or discharge demand. The project therefore develops a forecasting framework capable of estimating short-term care load and discharge demand while communicating uncertainty and supporting operational scenario analysis.

3. Objectives

The project objectives are to forecast the future number of children in HHS Care; develop a separate forecasting model for children discharged from HHS Care; compare assumed intake levels with predicted discharge demand; compare baseline, statistical, and machine-learning approaches; provide approximate prediction intervals; develop an interactive Streamlit dashboard; and provide forward-looking analytical information for planning.

4. Dataset & Preparation

The raw dataset contained 1,170 rows and six variables. After removing 450 completely blank rows, 720 recorded observations remained between January 12, 2023 and December 21, 2025. The expected calendar contained 1,075 days, leaving 355 missing dates. The data were converted into a continuous daily time series and missing numerical observations were interpolated. A was_missing indicator was retained, and no numerical missing values remained after preprocessing.

5. Methodology

The forecasting workflow consisted of data cleaning, date conversion, missing-date detection, continuous daily time-series construction, exploratory analysis, feature engineering, chronological train/test splitting, model training, evaluation, future forecasting, uncertainty estimation, scenario analysis, and dashboard presentation. Features included lag values at 1, 7, and 14 days; 7-day and 14-day rolling means and variances; a flow signal defined as Transfers minus Discharges; and day-of-week and month features. Rolling statistics used shifted historical observations to reduce target leakage.

6. Models Evaluated

The project evaluated Naive Persistence and 7-Day Moving Average as baseline methods; Exponential Smoothing and ARIMA (1,1,1) as statistical approaches; and Random Forest Regressor and Gradient Boosting Regressor as machine-learning approaches. A separate machine-learning pipeline was created for discharge demand forecasting.

7. Key HHS Care Model Results

Model	MAE	RMSE	MAPE
Naive Persistence	6.65	8.25	0.30%
7-Day Moving Average	21.56	26.58	0.96%
Exponential Smoothing	211.56	264.82	9.29%
ARIMA (1,1,1)	233.32	286.88	11.00%
Random Forest	39.29	61.26	1.86%
Gradient Boosting	54.25	76.37	2.57%

The Naive Persistence baseline had lower error in this particular evaluation setup, demonstrating why simple baselines are important. Some baseline/statistical and machine-learning evaluations used different forecasting procedures; standardized rolling-origin evaluation is identified as a future improvement.

8. Discharge Demand Results

A separate model was developed for children discharged from HHS Care. Random Forest achieved MAE 3.47, RMSE 4.67, and MAPE 30.79%. Gradient Boosting achieved MAE 4.43, RMSE 5.45, and MAPE 54.67%. Discharge MAPE is relatively high because small and zero discharge values can make percentage errors disproportionately large; therefore, MAE and RMSE are more useful for practical interpretation.

9. Forecast Results

The latest recorded HHS Care value was 2,484 children on December 21, 2025. The final system generated a 30-day forecast from December 22, 2025 through January 20, 2026. The predicted HHS Care averages were approximately 2,470.07 over 7 days, 2,473.96 over 14 days, and 2,480.30 over 30 days. The 30-day average predicted discharge demand was approximately 9.00 discharges per day.

10. Uncertainty & Scenario Analysis

Approximate residual-based prediction intervals were used to communicate uncertainty. The HHS Care residual standard deviation was approximately 55.75 with an approximate interval width of 109.28; the discharge residual standard deviation was approximately 4.62 with an approximate interval width of 9.06. These are residual-based approximate prediction intervals, not formal statistical confidence intervals.

Scenario	Assumed Intake	Avg. Daily Imbalance	30-Day Cumulative
Low Intake	7.56	-1.44	-43.27
Expected Intake	9.44	+0.45	+13.40
High Intake	11.33	+2.34	+70.07

The scenarios use assumed intake levels based on the recent 30-day average transfer level of approximately 9.44 per day. They are sensitivity-analysis assumptions, not direct predictions of future transfers.

11. Dashboard & Project Value

The Streamlit dashboard provides 7-, 14-, and 30-day forecast horizon selection; Random Forest and Gradient Boosting model selection; HHS Care and discharge forecasts; approximate uncertainty bands; model comparison; intake-versus-exit scenario analysis; and historical relative-load monitoring. Because the dataset contains no explicit operational capacity variable, the dashboard uses a historical 90th-percentile reference as a relative-load indicator rather than claiming actual capacity exceedance.

12. Limitations & Future Scope

Key limitations include 355 missing calendar dates requiring interpolation, the absence of an actual capacity variable, approximate residual-based prediction intervals, assumptions required for recursive future forecasting, unstable discharge MAPE for small or zero actual values, and differences in evaluation procedures across some model families. Future work can add capacity and operational variables, holiday information, automated retraining, probabilistic forecasting, additional forecasting models, and standardized rolling-origin validation.

13. Conclusion

The project demonstrates an end-to-end predictive forecasting framework for HHS Care load and discharge demand. By combining time-series analysis, machine learning, uncertainty estimation, and scenario analysis with an interactive Streamlit dashboard, the system provides forward-looking analytical information for operational planning while clearly communicating assumptions and data limitations.

Project Deliverables: Research Paper, Executive Summary, Google Colab Notebook, Streamlit Dashboard, Forecast Data Files, Model Evaluation Results, Scenario Analysis, GitHub Repository, README, Requirements File, Deployed Dashboard, and Feedback/Demonstration Video.